# Neural Network Field Guide

## 42 architectures and training schemes, from the perceptron (1958) to diffusion LLMs (2025).

Each diagram draws a network in miniature, from input at the bottom to output at the top. Most circles are neurons; in sequence models a column is a token, a circle its whole vector at one layer. A square is larger: an image patch, a matrix-valued state, a whole network, or an environment. Color and mark both show a cell's role, so diagrams read without color. Faded parts play little or no role in the step shown. Parts repeated over positions, windows, copies, or passes share weights; layers, experts, encoders, and decoders drawn alike do not. Sections group by a defining trait: wiring, dynamics, training, cost, or purpose. Years and training methods follow landmark papers, not always the first; many can be trained other ways.

**Input**
- Input unit
- Image patch
- Noisy input
- Random noise
- Visible unit (in and out)
- Masked token

**Hidden**
- Hidden unit
- Probabilistic hidden
- Capsule (vector output)
- Router

**State**
- Recurrent unit
- Memory cell
- Matrix-valued state

**Mixing**
- Attention
- Convolution or pooling

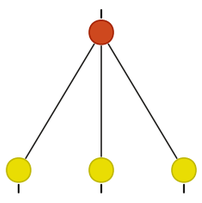
**Output**
- Output unit
- Reconstruction of input
- Velocity

**Whole systems**
- A whole network
- Environment

**Links**
- Connection
- Skip connection
- Data passed on or fed back
- Gate, acting at the dot
- Similarity score
- Learnable function

### FEED-FORWARD

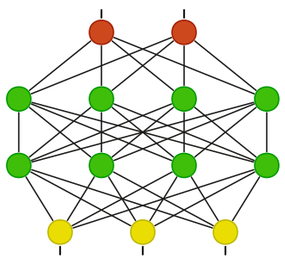
#### Perceptron



Weighted sum, then a threshold

`1958 · supervised`

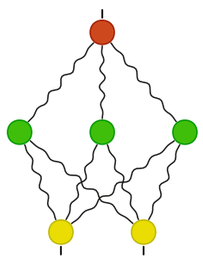
#### MLP



Stacked dense layers

`1986 · supervised`

#### KAN

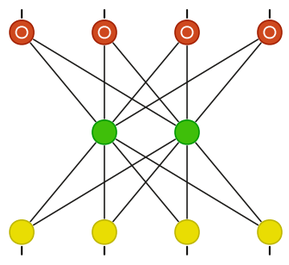


Learnable functions on edges

`2024 · supervised`

### AUTOENCODERS

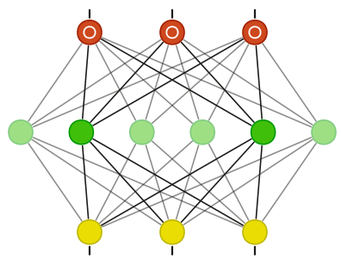
#### Autoencoder



Encode, then reconstruct

`1986 · reconstruct`

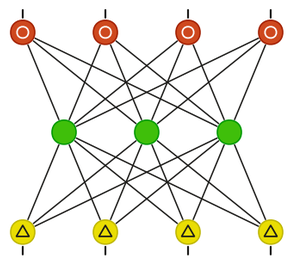
#### Sparse autoencoder



Few active units per input

`2006 · reconstruct`

#### Denoising autoencoder

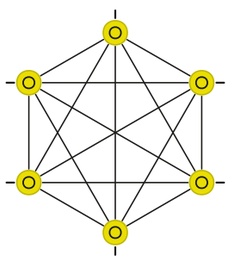


Clean up a noisy input

`2008 · denoise`

### ENERGY-BASED AND SELF-ORGANIZING

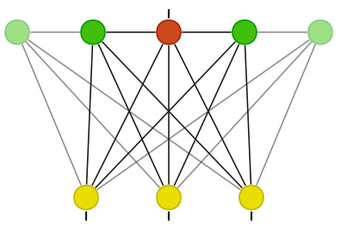
#### Hopfield network



Settles toward a stored pattern

`1982 · hebbian`

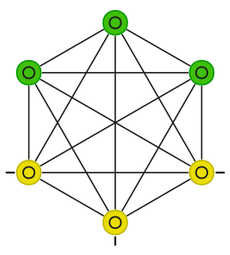
#### Kohonen map



Winner and neighbors move

`1982 · competitive`

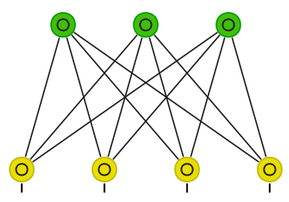
#### Boltzmann machine



Stochastic units, fully connected

`1985 · likelihood`

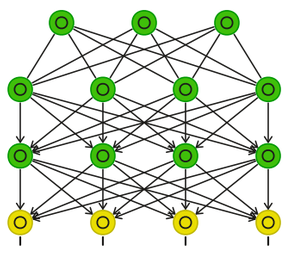
#### RBM



No links within a layer

`1986 · likelihood`

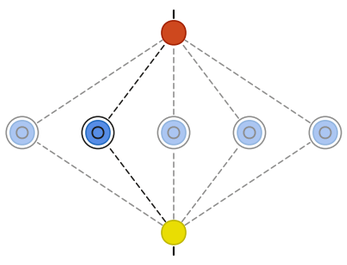
#### Deep belief network



Stacked RBMs, trained greedily

`2006 · likelihood`

#### Modern Hopfield network

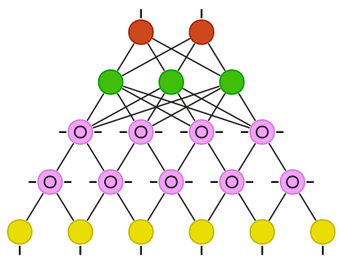


Attention as memory lookup

`2020 · supervised`

### CONVOLUTIONAL

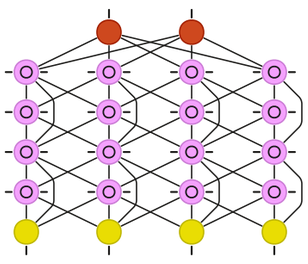
#### CNN



Local filters, shared weights

`1989 · supervised`

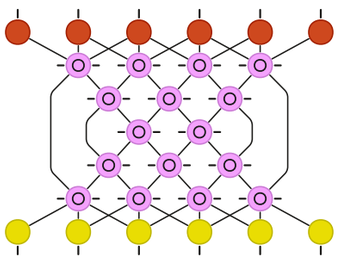
#### ResNet



Blocks add to a skip path

`2015 · supervised`

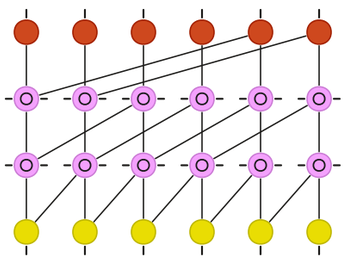
#### U-Net



Shrink, expand, skip across

`2015 · supervised`

#### WaveNet

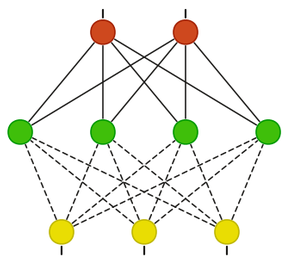


Dilated causal convolutions

`2016 · next token`

### SIMILARITY

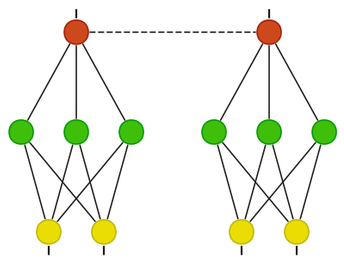
#### Radial basis network



Distance to stored centers

`1988 · supervised`

#### Siamese network

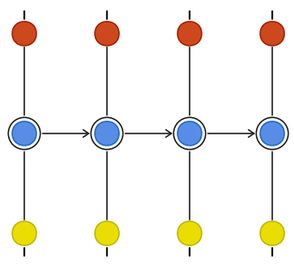


Twin towers, shared weights

`1993 · contrastive`

### RECURRENT AND MEMORY

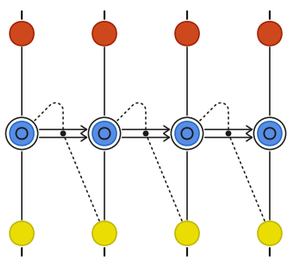
#### RNN



Hidden state passed along

`1990 · next token`

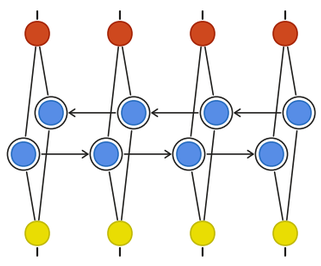
#### LSTM



Two states; forget gate from 1999

`1997 · next token`

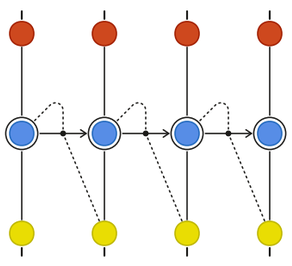
#### Bidirectional RNN



Reads the sequence both ways

`1997 · supervised`

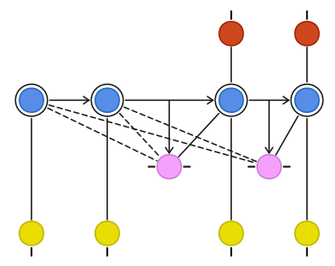
#### GRU



One state, two gates

`2014 · next token`

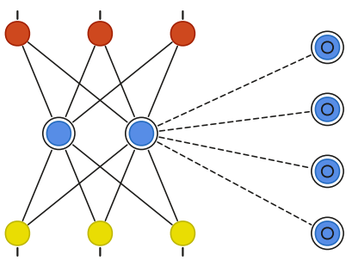
#### Seq2seq with attention



Learns to align and translate

`2014 · next token`

#### Neural Turing machine

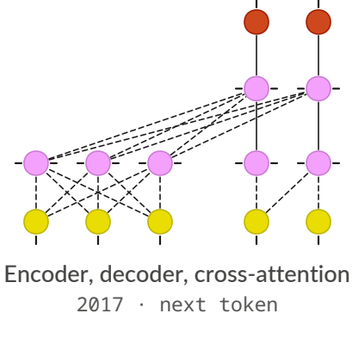


Controller reads, writes memory

`2014 · supervised`

### ATTENTION

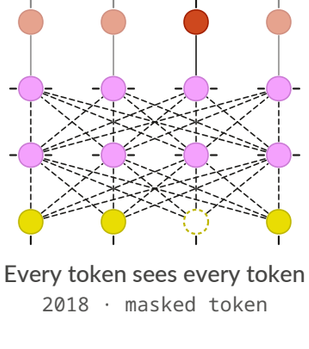
#### Transformer



Encoder, decoder, cross-attention

`2017 · next token`

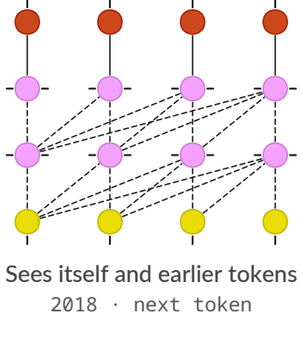
#### BERT



Every token sees every token

`2018 · masked token`

#### GPT-style LLM



Sees itself and earlier tokens

`2018 · next token`

#### ViT

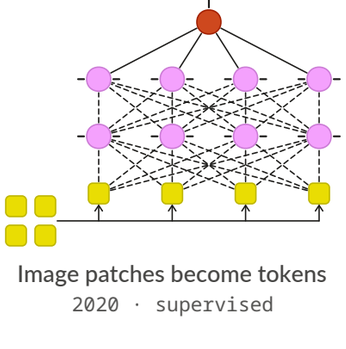
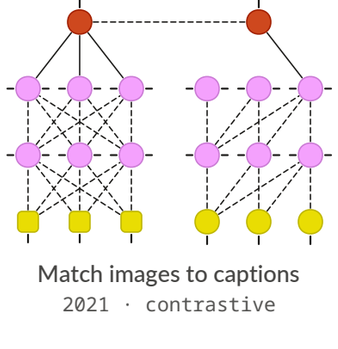


Image patches become tokens

`2020 · supervised`

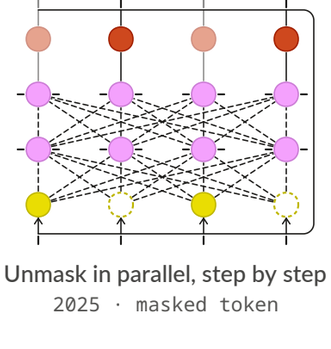
#### CLIP



Match images to captions

`2021 · contrastive`

#### Masked diffusion LLM

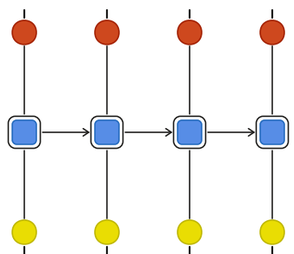


Unmask in parallel, step by step

`2025 · masked token`

### LINEAR-TIME MODELS AND HYBRIDS

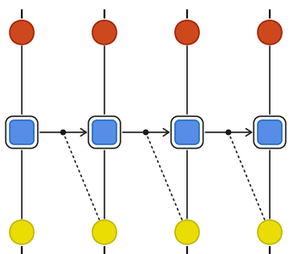
#### Linear attention



Softmax-free attention as an RNN

`2020 · next token`

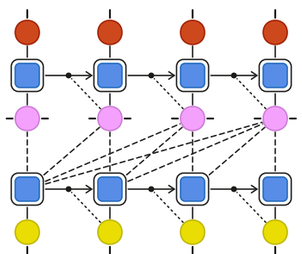
#### Mamba



Selective state, linear time

`2023 · next token`

#### Mamba–attention hybrid

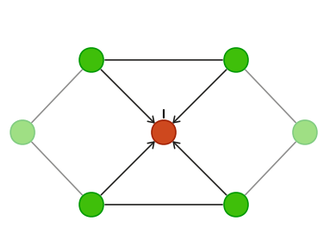


Mostly recurrent, some attention

`2024 · next token`

### ROUTING AND GRAPHS

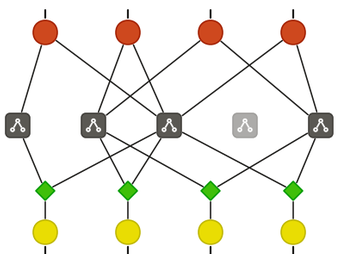
#### Graph neural network



Nodes pool neighbor messages

`2005 · supervised`

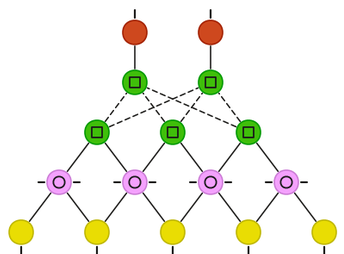
#### Sparse MoE



Each token → its top-*k* experts

`2017 · next token`

#### Capsule network

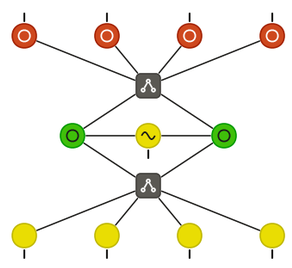


Vectors routed by agreement

`2017 · supervised`

### GENERATIVE FRAMEWORKS

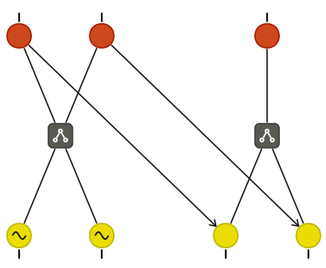
#### Variational autoencoder



Sample a code, then decode

`2013 · variational`

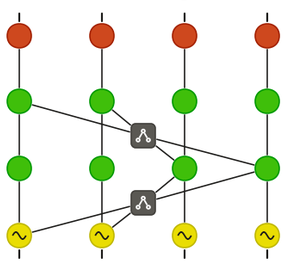
#### GAN



Generator vs. discriminator

`2014 · adversarial`

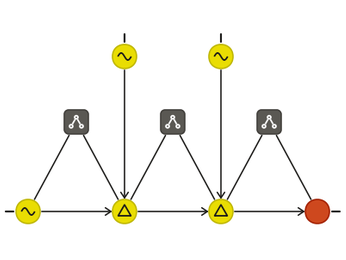
#### Normalizing flow



Invertible map from noise to data

`2014 · likelihood`

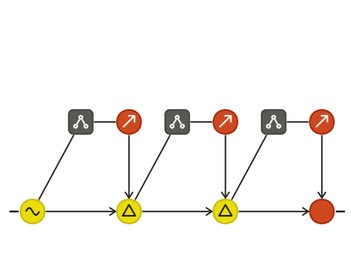
#### Diffusion model



One denoiser, run *T* times

`2015 · variational`

#### Flow matching

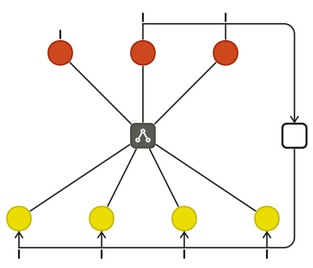


Learns a velocity, noise to data

`2022 · flow matching`

### REINFORCEMENT

#### Actor-critic (PPO)



More of what earned reward

`2017 · reward`

Inspired by The Neural Network Zoo by Fjodor van Veen (Asimov Institute, 2016; updated with Stefan Leijnen, 2019). www.asimovinstitute.org/neural-network-zoo

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